FINAL PROJECT REPORT

Windows Server & Active Directory Infrastructure Lab

System & Network Administration Laboratory Project

Project Title	Windows Server & Active Directory Infrastructure Lab
Operating System	Windows Server 2016
Virtualization Platform	VMware Workstation
Domain Name	lab.local
Domain Controller	DC01
Project Type	System & Network Administration Lab

1 Executive Summary

This project focused on the design, deployment, configuration, and administration of a Windows Server 2016-based Active Directory infrastructure in a virtualized laboratory environment.

The primary objective was to develop practical skills in Windows Server administration and enterprise identity management by deploying a Domain Controller, configuring Active Directory Domain Services (AD DS), integrating DNS, creating Organizational Units, managing users and security groups, and implementing Group Policy.

The project was implemented using VMware Workstation to create a controlled virtual laboratory environment. Windows Server 2016 was installed and configured as the primary Domain Controller named `DC01`. The server was promoted to a Domain Controller for the `lab.local` Active Directory domain.

The project also included structured user and group management, Organizational Unit design, security policy configuration, Group Policy implementation, service verification, Active Directory diagnostics, DNS testing, and troubleshooting documentation.

The Windows client domain-joining phase was not completed because a Windows client ISO was not available during the implementation of the project. The server-side Active Directory infrastructure was therefore completed and verified independently. The domain-joining phase remains a possible future extension of the laboratory.

2 Introduction

Modern organizations rely heavily on centralized identity and access management to control users, computers, resources, and security policies.

Microsoft Active Directory Domain Services provides a centralized platform for managing organizational identities and resources within a Windows-based enterprise environment.

This project was designed to simulate a small enterprise Windows Server infrastructure using VMware Workstation. The laboratory environment provided hands-on experience with the deployment and administration of Windows Server 2016 and Active Directory.

The project demonstrates practical knowledge of:

- Windows Server installation
- Server configuration
- Network configuration
- Active Directory Domain Services
- Domain Controller deployment
- DNS administration

- Organizational Unit management
- User account management
- Security group management
- Group Policy administration
- Active Directory troubleshooting
- PowerShell administration
- Domain Controller diagnostics

3 Project Objectives

The main objectives of this project were:

1. Install Windows Server 2016 in a virtualized environment.
2. Configure the Windows Server with an appropriate hostname and network settings.
3. Configure a static IP address for the server.
4. Install Active Directory Domain Services.
5. Promote the server to a Domain Controller.
6. Create and configure the `lab.local` Active Directory domain.
7. Configure and verify DNS services.
8. Create a structured Organizational Unit hierarchy.
9. Create and manage Active Directory users.
10. Create security groups for different departments.
11. Assign users to appropriate security groups.
12. Configure Group Policy Objects.
13. Apply basic security and user restrictions.
14. Verify Active Directory and DNS functionality.
15. Perform Domain Controller diagnostics.
16. Verify critical Windows Server services.
17. Develop troubleshooting procedures for common Active Directory problems.
18. Document the complete implementation for professional portfolio use.

4 Project Environment

The project was implemented in a virtualized laboratory environment.

Component	Configuration
Server Operating System	Windows Server 2016

Virtualization Platform	VMware Workstation
Server Name	DC01
Domain Name	lab.local
Server Role	Domain Controller
Directory Service	Active Directory Domain Services
DNS	Windows Server DNS
Network	Virtual VMware Network
Client Domain Joining	Not completed
Client ISO	Not available during implementation

5 Project Architecture

The laboratory architecture was designed around a Windows Server 2016 Domain Controller.

```
                    VMware Virtual Network
                              │
                              │
                              ▼
                  ┌──────────────────────┐
                  │         DC01         │
                  │  Windows Server 2016 │
                  │                      │
                  │  Active Directory    │
                  │  DNS Server          │
                  │  Domain Controller   │
                  └──────────────────────┘
                              │
                              ▼
                      Active Directory
                         lab.local
                              │
               ┌──────────────┼──────────────┐
               │              │              │
              IT             HR           Finance
               │              │              │
               └──────────────┼──────────────┘
                              │
                         Management
```

The Domain Controller `DC01` provides centralized Active Directory and DNS services for the `lab.local` domain.

6 Windows Server 2016 Installation

The first stage of the project involved creating a virtual machine using VMware Workstation.

The virtual machine was configured with appropriate hardware resources and Windows Server 2016 was installed successfully.

The installation process included:

- Virtual machine creation
- CPU and memory allocation
- Virtual disk configuration
- Network adapter configuration
- Windows Server 2016 installation
- Initial administrator configuration
- First system login

After installation, the server was prepared for further configuration.

7 Initial Server Configuration

After installing Windows Server 2016, the initial server configuration was performed.

The configuration included:

- Server hostname configuration
- Network adapter configuration
- Static IP address configuration
- Default gateway configuration
- DNS configuration
- Windows Server updates
- Firewall verification
- System configuration verification

The server was assigned the hostname:

```
DC01
```

The hostname was selected to clearly identify the system as the primary Domain Controller.

8 Network Configuration

A static IP configuration was applied to the Domain Controller.

The network configuration included:

```
IP Address:
[Configured DC01 IP]

Subnet Mask:
[Configured Subnet Mask]

Default Gateway:
[Configured Gateway]

Preferred DNS:
[Configured DNS Server]
```

The actual addressing information used in the lab was documented separately in the project configuration notes.

Network connectivity was verified using standard networking commands such as:

```
ipconfig /all
```

```
ping
```

and:

```
nslookup
```

These tests confirmed the basic network and name-resolution functionality required for Active Directory.

9 Active Directory Domain Services

Active Directory Domain Services was installed through Server Manager.

The server was then promoted to a Domain Controller.

The Active Directory domain created for the project was:

```
lab.local
```

The Domain Controller was:

```
DC01
```

The deployment established a centralized directory environment for managing users, groups, Organizational Units, and Group Policy.

The Active Directory structure was designed to simulate a small enterprise environment.

10 DNS Server Configuration

DNS is a critical dependency of Active Directory.

DNS services were configured on the Windows Server 2016 Domain Controller.

The DNS environment was configured to support the Active Directory domain:

```
lab.local
```

DNS verification was performed using:

```
nslookup dc01.lab.local
```

and:

```
nslookup lab.local
```

Additional DNS testing was performed using reverse lookup queries where applicable.

The DNS configuration was verified to ensure that the Domain Controller could resolve the required domain names.

11 Organizational Unit Structure

Organizational Units were created to provide logical structure and simplify administration.

The following Organizational Units were created:

```
lab.local
|
├── IT
├── HR
├── Finance
├── Management
├── Workstations
└── Security-Groups
```

The OU structure allows administrators to organize users, computers, and groups according to organizational requirements.

This structure also provides a foundation for applying Group Policy to specific departments or organizational units.

12 User Account Management

User accounts were created and organized according to departmental requirements.

The user structure included accounts for:

- IT department
- HR department
- Finance department
- Management department

Example structure:

```
IT
├── it.admin
└── it.user

HR
├── hr.manager
└── hr.user

Finance
├── finance.manager
└── finance.user

Management
└── manager.user
```

User accounts were configured through Active Directory Users and Computers.

The project demonstrated centralized user management and organizational separation using Active Directory.

13 Security Group Management

Security groups were created to simplify access management and policy administration.

The groups included:

```
IT-Admins
IT-Users
HR-Users
Finance-Users
Management-Users
```

Users were assigned to appropriate security groups based on their organizational roles.

The group-based management approach provides a scalable method for assigning permissions and applying security policies.

For example:

```
IT-Admins
└── it.admin

IT-Users
└── it.user

HR-Users
├── hr.manager
└── hr.user

Finance-Users
├── finance.manager
└── finance.user

Management-Users
└── manager.user
```

14 Group Policy Configuration

Group Policy was configured to demonstrate centralized management of security and user settings.

The following Group Policy Objects were created:

```
Domain-Security-Policy
Desktop-Restrictions
User-Restrictions
```

The policies were designed to demonstrate:

- Password security
- Account lockout configuration
- Desktop restrictions
- User restrictions
- Centralized policy management

Group Policy Management was used to create and manage the GPOs.

The project demonstrated the relationship between:

```
Active Directory
↓
Organizational Units
```

```
↓
Group Policy Objects
↓
Centralized Policy Management
```

15 Active Directory Verification

After completing the configuration, the Active Directory infrastructure was verified.

The following commands and tools were used:

```
dcdiag
nslookup
ipconfig /all
ping
```

PowerShell commands were also used:

```
Get-ADDomain
Get-ADForest
Get-ADDomainController
Get-ADUser -Filter *
Get-ADGroup -Filter *
Get-ADOrganizationalUnit -Filter *
```

These tests were used to verify the Active Directory domain, Domain Controller, users, groups, and Organizational Units.

16 Windows Server Services Verification

Critical services required for Active Directory functionality were verified.

The following services were checked:

```
NTDS
DNS
Netlogon
KDC
Windows Time
```

PowerShell was used to verify service status:

```
Get-Service NTDS,DNS,Netlogon,KDC,W32Time
```

The verification process confirmed that the required services were configured and available for the Active Directory environment.

17 Troubleshooting

A dedicated troubleshooting phase was included in the project.

The troubleshooting documentation covered:

- DNS troubleshooting
- Active Directory troubleshooting
- Domain Controller diagnostics
- Group Policy troubleshooting
- Service verification
- Event Viewer investigation

The following tools were used or documented for troubleshooting:

```
ipconfig
ping
nslookup
dcdiag
gpupdate
gpresult
PowerShell
Event Viewer
DNS Manager
Active Directory Users and Computers
Group Policy Management
Services Console
```

A structured troubleshooting methodology was followed:

```
Identify Problem → Collect Information → Check Configuration → Test Services →
Identify Root Cause → Apply Solution → Verify Resolution
```

The troubleshooting documentation is available in the project's `13-Troubleshooting` directory.

18 Testing Results

The following components were successfully configured and verified:

Test Area	Result

Windows Server 2016 Installation	**Completed**
Server Hostname Configuration	**Completed**
Static IP Configuration	**Completed**
Active Directory Domain Services	**Completed**
Domain Controller Deployment	**Completed**
lab.local Domain	**Completed**
DNS Configuration	**Completed**
DNS Resolution Testing	**Completed**
Organizational Units	**Completed**
User Accounts	**Completed**
Security Groups	**Completed**
Group Membership	**Completed**
Group Policy Configuration	**Completed**
Domain Controller Diagnostics	**Performed**
Windows Server Services Verification	**Performed**
Troubleshooting Documentation	**Completed**
Windows Client Domain Joining	**Not Completed**

19 Limitations

The primary limitation of this project was the lack of a Windows 10/11 Pro client ISO.

As a result, the following activities were not completed:

- Windows client virtual machine deployment
- Windows client domain joining
- Domain user login from a Windows client
- Client-side Group Policy verification
- Client-side domain authentication testing

These activities can be added as a future extension by deploying a Windows 10/11 Pro virtual machine on the same VMware virtual network.

The server-side Active Directory infrastructure was completed independently and verified using the available Windows Server tools and administrative utilities.

20 Future Improvements

The following improvements can be added to expand the project:

1. Deploy a Windows 10/11 Pro client VM.
2. Join the client to the `lab.local` domain.
3. Test domain authentication using Active Directory users.
4. Verify Group Policy application on the client.
5. Add additional Domain Controllers for redundancy.
6. Configure Active Directory replication.
7. Implement centralized file services.
8. Integrate Linux clients with Active Directory.
9. Configure Samba-based file sharing.
10. Implement backup and recovery procedures.
11. Configure Windows Server monitoring.
12. Integrate the infrastructure with a centralized monitoring platform.

21 Skills Demonstrated

This project demonstrates practical knowledge of:

Windows Server Administration

- Windows Server 2016 installation
- Server configuration
- Network configuration
- Windows services
- Server troubleshooting

Active Directory

- AD DS installation
- Domain Controller deployment
- Domain management
- Organizational Unit design
- User management
- Security group management

DNS

- DNS configuration

- Forward lookup testing
- Reverse lookup testing
- DNS troubleshooting

Group Policy

- GPO creation
- GPO management
- Security policy configuration
- User restrictions
- Desktop restrictions

Troubleshooting

- Domain Controller diagnostics
- DNS troubleshooting
- Active Directory troubleshooting
- Group Policy troubleshooting
- Event Viewer analysis

PowerShell

- Active Directory queries
- Service verification
- Domain information gathering
- User and group verification
- Organizational Unit verification

22 Project Outcomes

The project successfully provided practical experience in deploying and administering a Windows Server 2016 Active Directory environment.

The completed laboratory demonstrates the ability to:

- Deploy a Windows Server environment.
- Configure a Domain Controller.
- Build an Active Directory domain.
- Configure DNS for Active Directory.
- Organize enterprise users through OUs.
- Manage users and security groups.
- Apply centralized Group Policy.

- Verify Domain Controller health.
- Troubleshoot common infrastructure issues.
- Use PowerShell for administrative verification.

The project provides a strong foundation for further enterprise infrastructure work and demonstrates practical System and Network Administration capabilities.

23 Conclusion

The Windows Server & Active Directory Infrastructure Lab was successfully implemented using Windows Server 2016 in a VMware Workstation virtualized environment.

The project covered the complete server-side deployment of an Active Directory infrastructure, beginning with Windows Server installation and initial configuration and continuing through Domain Controller deployment, DNS configuration, Organizational Unit creation, user and group management, Group Policy implementation, system verification, and troubleshooting.

The project provided hands-on experience with technologies and administrative practices commonly used in enterprise Windows environments.

Although Windows client domain joining could not be completed due to the unavailability of a Windows client ISO, the core server-side infrastructure was successfully established and verified.

This project serves as a practical demonstration of Windows Server administration, Active Directory management, DNS administration, Group Policy management, PowerShell usage, and infrastructure troubleshooting.

> The completed project forms a strong foundation for the next stage of the portfolio: **Enterprise Linux & Windows Integration**, where Windows Server Active Directory can be integrated with Linux systems and enterprise file-sharing services.